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14

TWO-DIMENSIONAL FLOWS

14.1 Grids for DSMC computations

The basic simulation procedures that were developed in Chapter 11 for homogeneous gases, and were applied in Chapter 12 to one-dimensional flows, are applicable also to two- and three-dimensional flows. It is the requirement for efficient schemes for the generation and utilization of grid systems that represents the new challenge when the direct simulation Monte Carlo (DSMC) method is applied to multi-dimensional problems. The ideal DSMC grid would:

(a) have high computational efficiency;

(b) allow efficient definition of complex flow geometries;

(c) be 'fitted' to any body and other significant flow boundaries;

(d) have variable cell sizes related to the local gradients in the flow properties and to the mean free path;

(e) allow this size to adapt to local conditions as the flow develops; and

(f) employ sub-cells for effective 'nearest neighbour' collision pair selection.

While some of these criteria are common to the grid generation problem for applications of the finite difference and finite element methods to continuum flows, there are basic differences in the requirements of the DSMC method. Continuum methods generally require a near orthogonal grid structure with no discontinuities in the lines that define this structure. The DSMC method uses the grid or cell system only for the sampling of the macroscopic properties and for the selection of possible collision partners. The cell structure can therefore be irregular with regard to the shape of the cells and there can be discontinuities in the grid lines that define the cells. However, if flow contours are to be constructed, it is generally desirable for the grid to be continuous.

The sampled density is used in the DSMC procedures for establishing the collision rate and, in order for this rate to realistic, it is desirable to have the number of molecules as large as possible, the desirable minimum being about ten to twenty. On the other hand, in the selection of possible collision partners, it is desirable to have this number as small as possible in order to reduce the mean spacing of collision pairs and thereby minimize the smearing of gradients. As in the one-dimensional programs, these conflicting requirements can been reconciled by dividing the sampling cell structure into a set of sub-cells for the selection of collision pairs.

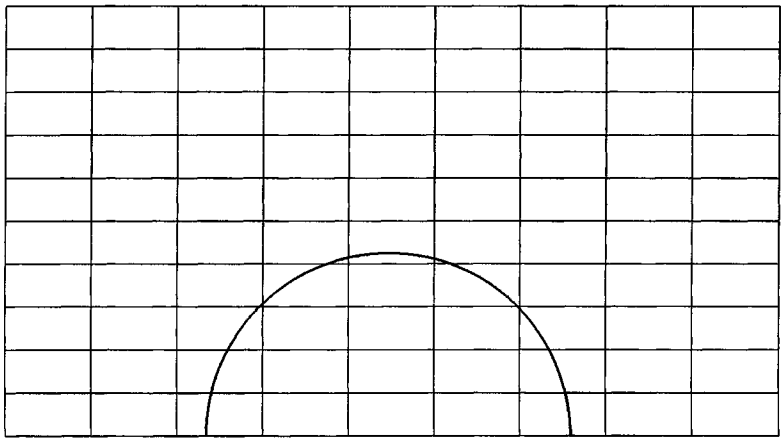


Fig. 14.1 Simple rectangular grid for the flow past a cylinder.

The existing cell schemes may be summarized as follows:

(i) *Simple constant area cells defined by a rectangular grid.*

These are almost always used in the first set of simulation programs by any group because of the ease of implementation. The scheme is very efficient computationally but has severe disadvantages. For flows with large density variations, some cells will contain far more than the optimum number of simulated molecules, while others will contain far fewer than the optimum number. More seriously, if surface boundaries are not rectangular as in the example in Fig. 14.1, the effective cells near the boundaries are irregular and the scheme is not adequate for serious calculations at low Knudsen numbers. Also, because the cell structure is unrelated to the surface geometry, each different surface requires special coding within the program.

(ii) *Multi-level rectangular grid.*

This feature is illustrated in Fig. 14.2 and alleviates some of the disadvantages of the simple scheme in which all cells are of the same size. The blocks of subdivided cells must be rectangular and it is not easy to define them efficiently when the flow geometry is complex. Changes in the flow geometry still require modifications to the program, and the scheme is not suited to general programs that accommodate new geometries through changes to the data alone.

(iii) *Adaptive rectangular cells.*

Instead of a whole block of cells being subdivided as in scheme (ii), individual cells may be subdivided when the density exceeds a set value. The grid can therefore be made to adapt to the flow and the scheme has been implemented by Babovsky et al. (1989).

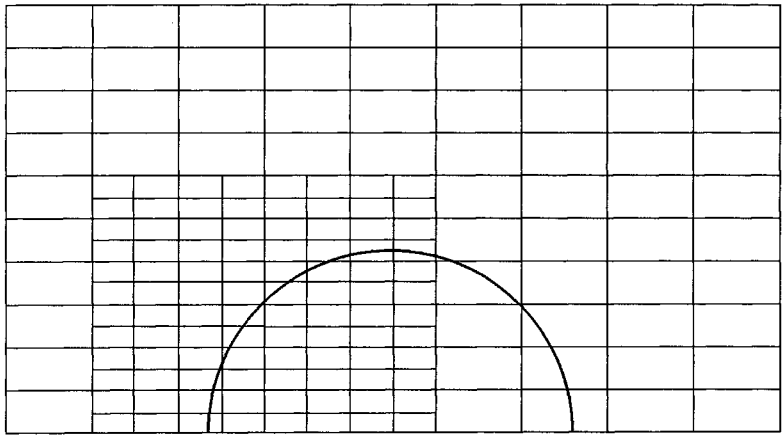


Fig. 14.2 Subdivided cells in a region of higher density.

(iv) *'Point reference' scheme.*

This scheme (Bird 1976) indexes the simulated molecules to a series of points rather than volumetric cells. The molecule is 'in a cell' when it is nearest to the point that defines that cell. This procedure was used in the first 'general' program to cope with arbitrary flow geometry in both two and three dimensions. However, there was an effective distortion of the flow boundaries for non-rectangular geometries and the cell volume could not be precisely defined. This led to serious difficulties with some flows and the method is largely of historical interest.

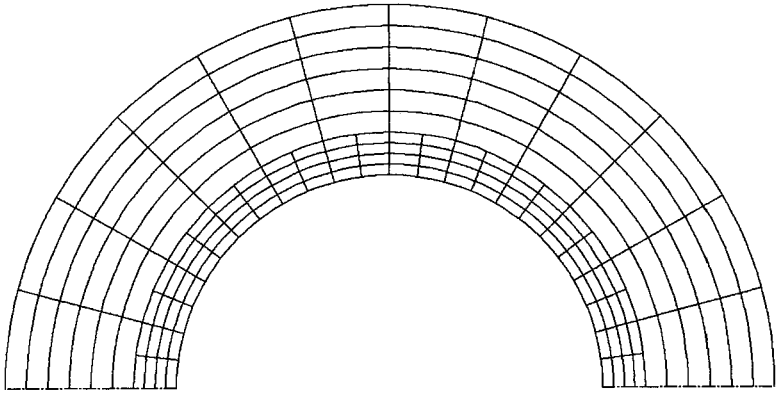


Fig. 14.3 Body fitted cells for a cylinder.

(v) *Body-fitted 'analytical' cells.*

This option is illustrated in Fig. 14.3 for the flow past a cylinder. One set of cell boundaries is geometrically similar to one of the boundaries (generally a body surface) that primarily defines the flow. The method can be used only when all the boundaries can be described analytically. Quadric surface geometries are particularly amenable to this approach because of the existence of the theorem of eqns (7.85)–(7.87). Given a specified initial location and direction cosines, this provides a simple quadratic expression for the distances to any intersections with the surface. New programs are generally required when there is a significant change in the geometry, and the programs are much less efficient than those with rectangular cells. On the other hand, the computational requirements for accuracy at low Knudsen numbers are readily satisfied by this scheme.

(vi) *'General' scheme.*

The flowfield is divided into an arbitrary number of four-sided regions. Two opposite sides are curved and the other two are straight. The curved sides are specified by two sets of points that may be specified individually or may lie on segments of straight lines and conic sections. Corresponding points are joined by straight lines and these are, in turn, divided into segments and the ends of these segments are joined to form the system of quadrilateral cells. Each of the four sides can be an axis, a plane of symmetry, an interface with a specified uniform or non-uniform flow, a solid surface which may be absorbing or outgassing, a vacuum, or an interface with another side. Virtually any flow can be constructed from these regions. The current implementation of this scheme by the author is called the 'G2' set of programs.

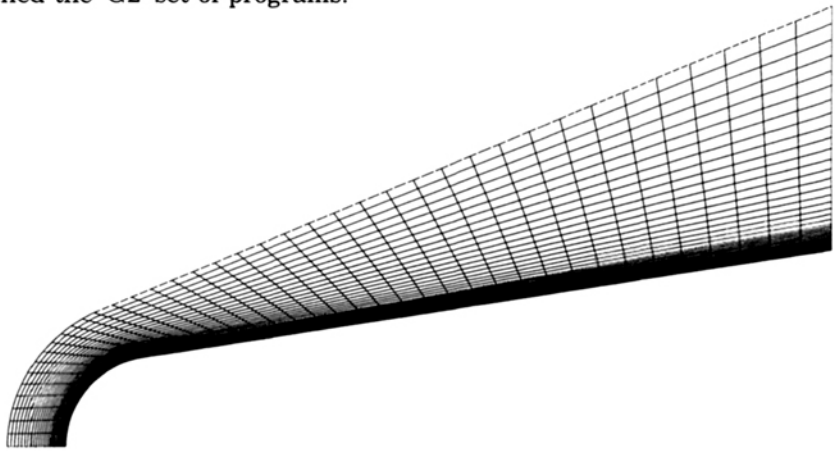


Fig. 14.4 Cells for blunt cone generated by the G2 program system.

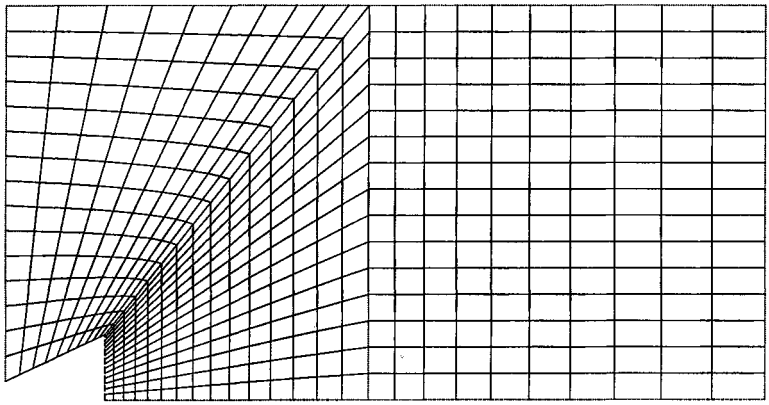


Fig. 14.5 Cells generated for a plume backflow calculation.

As shown in Figs 14.4 and 14.5, diverse problems can be accommodated without any changes being made to the code, but this is at the cost of computational efficiency. The molecules are traced through the grid by computing the intersections with all the cell boundaries. The computation time for the 'molecule movement' section of the programs is an order of magnitude larger than for a simple rectangular grid. The overall performance penalty depends on the Knudsen number and the consequent balance between the computation time spent in the move and collision routines, but there is typically a factor of five increase in the CPU time. As noted earlier, sub-cells are used to control the maximum separation of the collision partners and are highly desirable in regions of high vorticity. They are incorporated into the G2 programs but the smaller the sub-cells, the larger the computation time penalty.

(vii) *Orthogonal transformations to rectangular grids.*

This approach permits a rectangular grid to be used for flows that have curved boundaries in physical space. There are some restrictions on the geometries that allow transformation, and the necessity of moving the molecules along curved paths in the transformed space may cost more efficiency than is gained by the rectangular grid.

(viii) *Rectangular multi-level sub-cells with adaptive body-fitted cells.*

This is based on a fixed rectangular grid of very small sub-cells. The body intersects these and is specified by the lines joining its points of intersection with the sub-cell boundaries. The sub-cells that contain a surface intersection are regarded as zero level cells. Those within the flow that touch these are 'level 1' cells and so on throughout the flow. These sub-cell levels provide an easy determination of whether possible surface interactions need be considered when molecules are moved. When these need not be considered, the procedures are as fast as in the programs

described in scheme (i). The initial cells are simple rectangular cells as shown in Fig. 14.1. Once a steady flow has been established, an automatic adaptive re-gridding can take place. The resulting grid structure for the two-dimensional flow past a circular cylinder is shown in Fig. 14.6.

The cells in Fig. 14.6 may look bizarre in comparison with those for continuum calculations, but the DSMC method uses the grid or cell system only for the sampling of the macroscopic properties and for the establishment of the collision rate, with collision partners selected from the sub-cells. The cell structure can therefore be irregular with regard to the shape of the cells and there can be discontinuities in the grid lines that define the cells. The sub-cells may be sub-divided in the vicinity of the surface so that the separation of the collision partners is reduced to the values that are required for accurate results at low Knudsen numbers. The time step and the number of real molecules represented by each simulated molecule may be reduced in cells near the surface in order to establish a reasonable average number of molecules in the very small sub-cells. While the boundaries of the sub-cell region is rectangular, the boundary of the simulation region may be inset from these boundaries in order to avoid the computation of unnecessarily large regions of uniform flow.

There is generally a trade-off between the ease of setting up a case with complex boundaries and the computational efficiency of the programs. The ideal program would take care of all the computational details and would require the user to specify only the geometry and the undisturbed flow conditions.

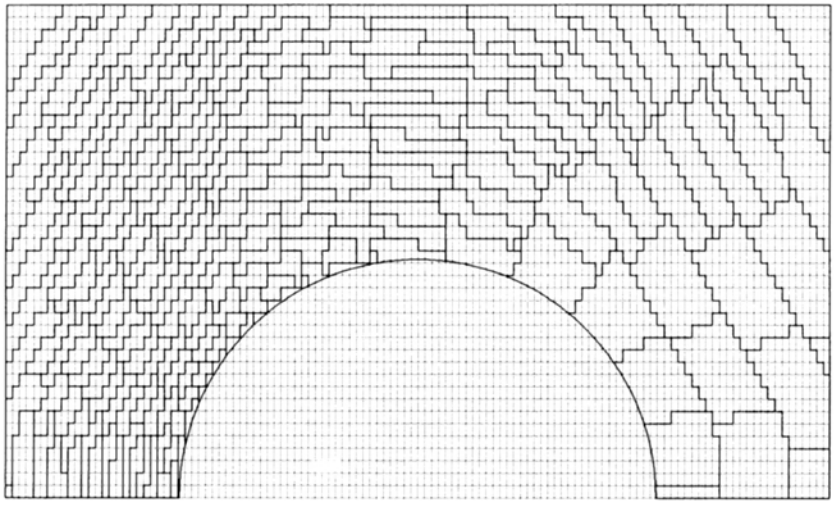


Fig. 14.6 An automatically generated adaptive grid for the flow past a circular cylinder.

14.2 DSMC program for two-dimensional flows

The demonstration program DSMC2.FOR deals only with problems involving flat surfaces that can lie along the cell boundaries of a simple rectangular grid. The flowfield is rectangular and is aligned such that the sides are parallel to the x-and y-axes. Each of the four sides may be a boundary with a uniform stream, a plane of symmetry, or a vacuum. The stream is specified by the velocity components in the x-and y-directions, the number density, the temperature, and the composition. The flowfield is divided into a grid of rectangular cells. As with program DSMC1, the cell spacing may be either uniform or in a geometric progression. There may be one or two single-sided surfaces that must be parallel to either the x- or y-axis, lie along a cell boundary, and terminate at one of the grid points. The size and locations of the surfaces are specified through these cell boundaries and grid points. The two single-sided surfaces may be placed back-to-back to form a double-sided surface. The surface temperatures are assumed to be uniform and it is assumed that the reflection is diffuse with complete thermal accommodation. In addition to the surfaces, the flow may include a jet. The plane of the jet efflux is specified in a similar manner to a surface and the initial direction of the jet must be aligned with one of the axes. The efflux is uniform and is specified by its number density, temperature, and composition. The specification of the gas is similar to that in program DSMC1.

14.3 The supersonic leading-edge problem

The 'leading-edge' or 'horizontal flat plate' problem relates to the two-dimensional flow past a semi-infinite thin flat plate that is aligned with the stream. It is of fundamental interest in gas dynamics and has been the subject of many theoretical and experimental studies. While continuum theory is frequently applied to this problem, as in the Blasius theory for the incompressible laminar boundary layer, the solutions are invalid very close to the edge of the plate. The unresolvable leading-edge singularity in continuum gas dynamics becomes a well-defined problem in molecular gas dynamics.

In the case of a supersonic external flow, the displacement effect of the boundary layer produces a weak oblique shock wave in the external flow. At a distance of some tens of mean free paths from the leading edge, the shock wave structure overlaps with that of the boundary layer and, as the leading edge is approached, the shock and boundary layer gradually merge and lose their separate identities. Even in a hypersonic stream, the disturbance extends a few mean free paths upstream of the leading edge and, contrary to some descriptions, the conditions at the leading edge do not correspond to those of free molecule flow. However, the velocity slip at the leading edge is a large fraction of the freestream velocity. This slip velocity gradually decreases along the plate but, again contrary to the continuum assumption, it does not disappear completely.

The supersonic leading-edge problem has provided one of the most useful test cases for the verification of the DSMC method through comparison with experiment (e.g. Harvey 1986). One of its advantages as a test case is that the pressure disturbance is generated indirectly rather than through direct impact of the stream. In addition, the shear stress is of the same order as the pressure and this, together with the heat transfer, is particularly sensitive to any failure to meet the computational requirements of the DSMC method. This flow is therefore employed in §14.5 for a study of the effects of the cell size and time step in a DSMC computation.

The listed data in DSMC2.FOR is for the flow of nitrogen at a Mach number of four past a flat plate with a length of approximately 70 times the mean free path in the undisturbed gas. The temperature of the plate is 1.6 times the undisturbed gas temperature. The stream is in the positive x-direction and the plate lies along the lower y-boundary. The leading edge is 5.4 mean free paths from the upstream boundary and the plate extends to the downstream boundary. The plane in which the plate lies is specified as a plane of symmetry and the leading edge is effectively a transition from specular reflection to diffuse reflection. The downstream boundary is specified as a stream boundary but, at a Mach number of four, there is no practical difference between a stream boundary and a vacuum boundary. This means there is an interaction between the flow and the downstream boundary. Calculations were made for several plate lengths in order to determine the extent of the interference region from this unrealistic boundary condition. The grid is 100×60 and the cell widths and heights are both in geometric progression with an increase by a factor of two in both directions. There are two sub-cells per cell in each direction so that the computational requirements are satisfied by a narrow margin.

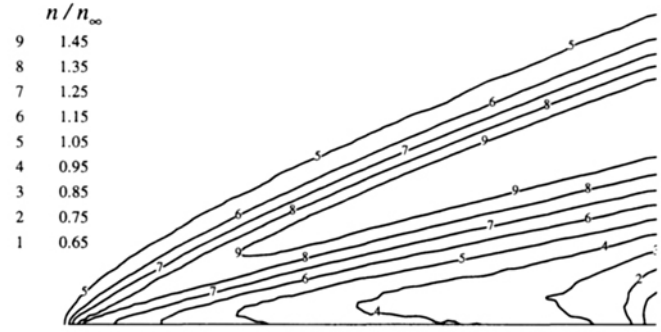


Fig. 14.7 Number density contours at $(Ma) = 4$ and $(Kn) = 0.0143$.

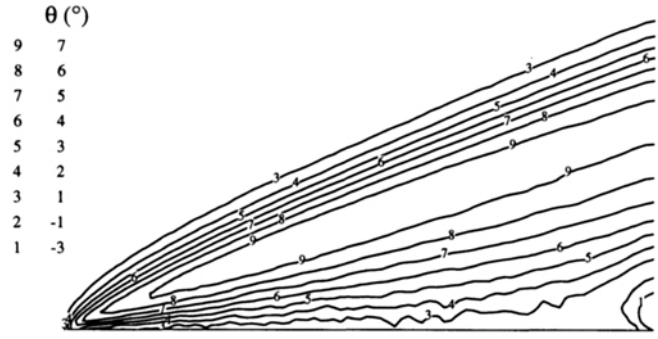


Fig. 14.8 Contours of constant flow angle.

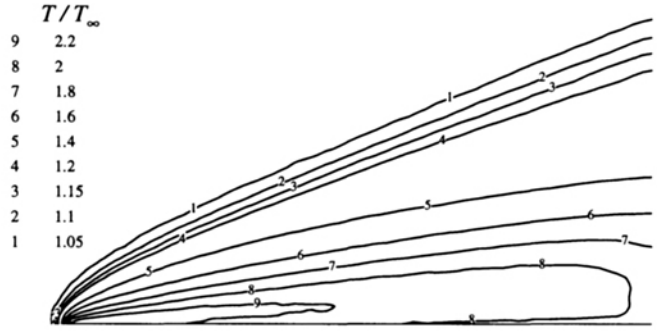


Fig. 14.9 Temperature contours at $(Ma) = 4$ and $(Kn) = 0.0143$.

The contours of constant number density are shown in Fig. 14.7. These show that the oblique shock wave and the boundary layer emerge as distinct identities at about 20 mean free paths from the leading edge. The flow angle contours of Fig. 14.8 indicate that the displacement effect of the boundary layer is equivalent to a wedge angle of just over seven degrees, and the strength of the oblique shock wave is consistent with this deflection angle. The number density increases across the shock, but it then decreases to a value near the surface that is less than the stream density. Fig. 14.9 shows that this low density is associated with temperatures that are higher than both the temperature at the rear of the shock and the surface temperature.

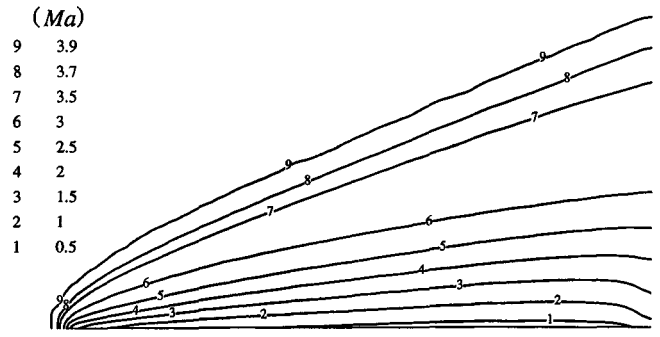


Fig. 14.10 Contours of constant value of the local Mach number.

The temperature contours show most clearly the extent of the flow disturbance upstream of the leading edge. The upstream temperature rise is approximately equal to that rise across the oblique shock. On the other hand, the upstream density increase is small compared with that across the shock. The final set of contours in Fig. 14.10 are for the local Mach number. These show the magnitude of the velocity slip at the surface in that Mach numbers less than 0.5 are confined to a thin region over the latter half of the plate. The ratio of the surface temperature to the stream temperature is 1.66 and the temperature contours show that the temperature slip is also substantial.

The unrealistic downstream boundary condition has a noticeable effect on the flow near the surface for at least ten to fifteen mean free paths upstream of this boundary. The vacuum boundary has a negligible effect when the Mach number based on the velocity component normal to the boundary is greater than about two. This means that the effect is most pronounced near the surface. There is a sharp fall in the density near the downstream end of the plate and the flow acquires a negative flow angle that reaches six degrees at the end of the plate. In addition, the region of subsonic flow becomes very narrow at the downstream boundary. The extent of the upstream influence from this boundary is more precisely defined by the surface properties that are plotted in Figs 14.11 to 14.14.

The number flux to the surface is compared to the free-molecule value in Fig. 14.11. The flux at the leading edge is substantially above the free-molecule value, but there is then a sharp increase in the ratio which reaches a maximum at about four or five mean free paths from the leading edge. This is followed by a gradual decline in the flux towards the free-molecule value, but this is concluded by a sharp decrease due to the downstream boundary condition.

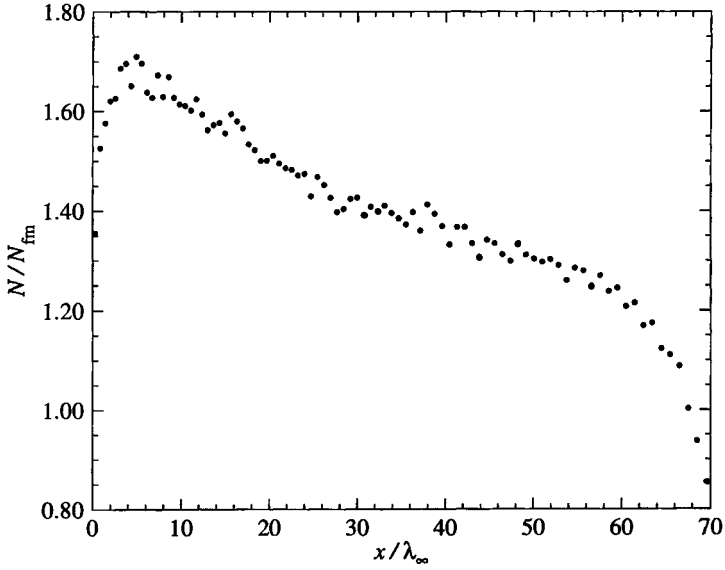


Fig. 14.11 The number flux as a ratio to the free-molecule value.

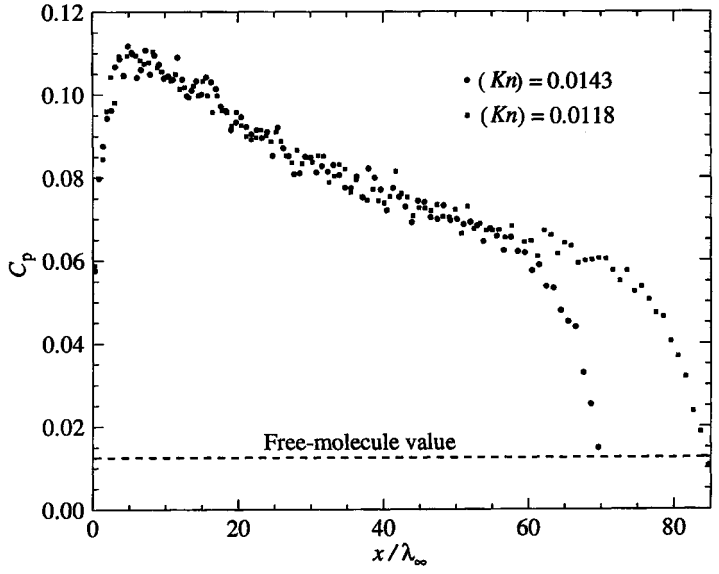


Fig.14.12 The pressure distribution along the plate.

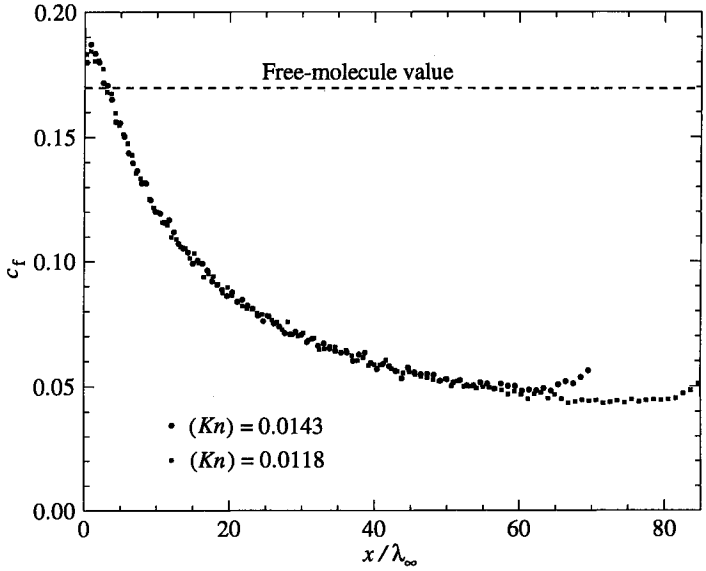


Fig. 14.13 The skin friction coefficient along the plate.

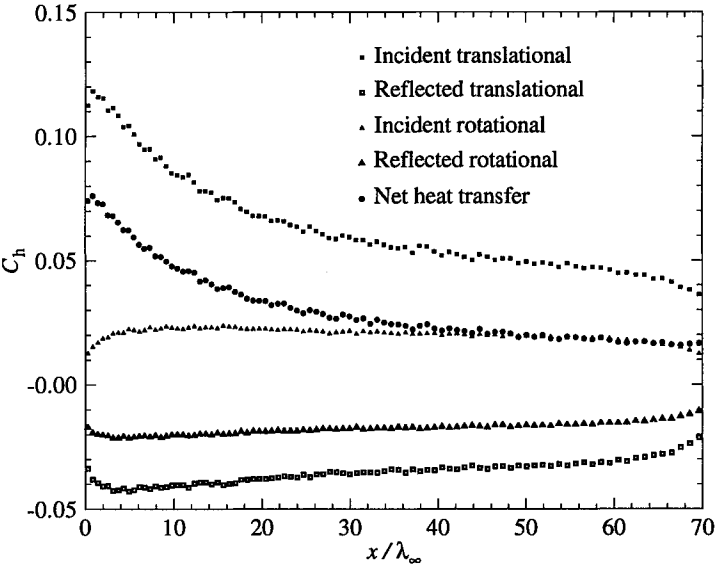


Fig. 14.14 The heat transfer coefficients along the plate.

In order to clearly define this influence, a similar run was made for a plate length of 85 freestream mean free paths. The pressure and shear stress distributions are shown for both calculations and it is clear that the upstream influence of the downstream boundary is largely confined to about 15 freestream mean free paths.

As expected, the pressure coefficient at the leading edge is well above the free-molecule value. The general shape of the pressure profile is similar to that for the number flux. The skin friction distribution is qualitatively different, in that the maximum occurs at the leading edge and is only slightly higher than the free-molecule value. The skin friction coefficient then falls well below the free-molecule value. Although the large pressure effect is inconsistent with laminar boundary layer theory, the shear stress is generally consistent with this theory in that the skin friction coefficient is very nearly inversely proportional to the square root of the distance from the leading edge. The acceleration of the flow near the surface to the downstream vacuum boundary causes an increase in the skin friction.

The heat transfer coefficient C_h is defined as the ratio of the heat flux per unit area to $\frac{1}{2}\rho u_\infty^3$. Separate results are shown in Fig. 14.14 for the translational and rotational contributions from both the incident and reflected molecules. The distribution of the incident translational heat flux is generally similar to the skin friction profile except that the boundary disturbance is a decrease rather than an increase. The surface temperature is uniform, so that the reflected distributions of both the rotational and translational components are similar to the number flux distribution. The incident and reflected components for the rotational energy are very nearly in balance and the internal mode has very little effect on the net heat flux. The net heat transfer coefficient is of the order of half the skin friction coefficient

14.4 Effect of cell size

The calculations in §14.3 were made with a cell grid that met the general requirements that were discussed in §10.5, although not by a large margin. The time step does not affect the results unless it is outside the recommended range by an unrealistically large amount. The cell size is most critical in the direction with the highest gradients and, for the flat-plate flow, this is the direction normal to the plate. The calculation in the previous section was repeated for a shorter plate of approximately 40 freestream mean free paths and with a uniform cell spacing. Runs were made for a number of cell and sub-cell arrangements in the normal (y) direction, while the cell size in the streamwise (x) direction remained fixed.

The first case with 40 cells and 80 sub-cells in the y-direction was expected to have an accuracy similar to the previous calculations. The resulting distributions of pressure coefficient and skin friction coefficient are shown in Figs 14.15 and 14.16, respectively. A comparison of these results with Figs 14.12 and 14.13 shows that, allowing for the effects due to the downstream boundary condition, the results are equivalent.

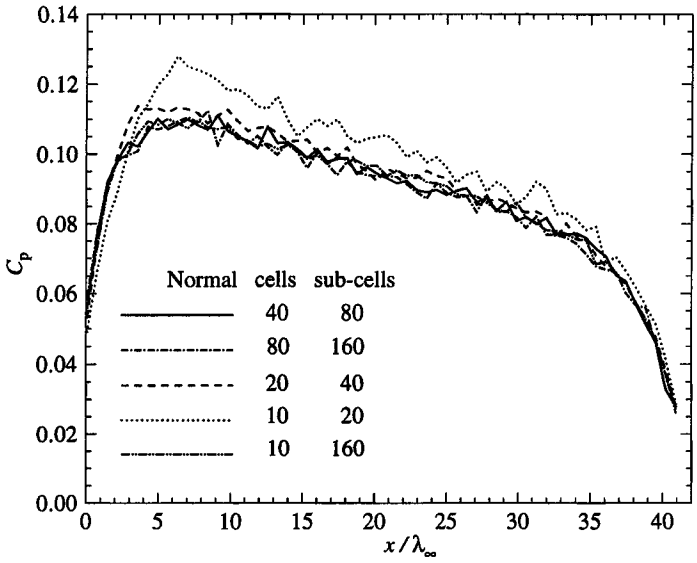


Fig. 14.15 The effect of cell size on the pressure distribution.

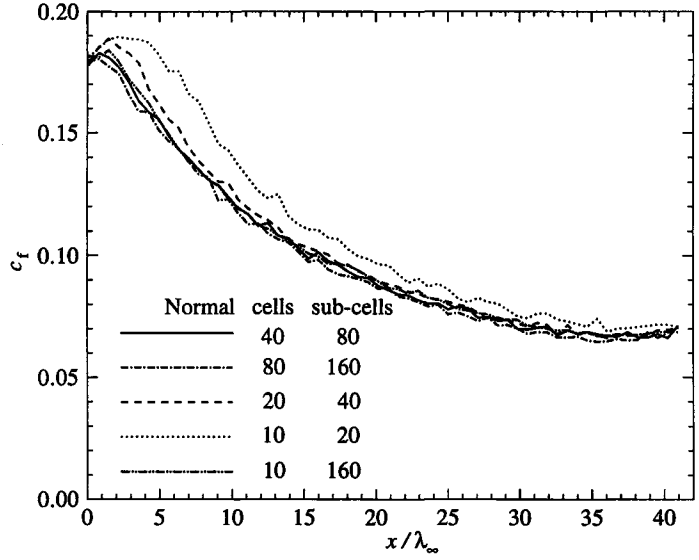


Fig. 14.16 The effect of cell size on the skin friction coefficient.